

Supplementary Materials for

Enhancing High-density Spatial Transcriptomics from Histology Images using HisHRST

Supplementary Table

Table S1. Comparison of HisHRST and baseline methods on tissue slices TENX116, TENX126, and TENX140 from the Xenium platform within the HEST-1k dataset.

Methods	TENX116			TENX126			TENX140		
	PCC	MSE	MAE	PCC	MSE	MAE	PCC	MSE	MAE
STNet [1]	0.134	0.985	0.678	0.124	0.990	0.624	0.163	1.316	0.756
DeepSpace [2]	0.298	0.857	0.569	0.316	0.915	0.537	0.353	1.017	0.613
HisToGene [3]	0.387	0.839	0.488	0.469	0.901	0.505	0.492	0.974	0.581
THItoGene [4]	0.457	0.714	0.421	0.519	0.720	0.496	0.534	0.987	0.568
STAGE [5]	0.495	1.317	0.544	0.587	0.629	0.378	0.677	0.748	0.409
HisHRST(ours)	0.532	0.783	0.395	0.742	0.383	0.259	0.775	0.441	0.256

Table S2. Summary of differentially expressed genes: cancer epithelial cells vs. other cell types. This table presents a subset of the significantly upregulated genes identified in cancer epithelial cells based on the Wilcoxon rank-sum test. Shown are the gene names, adjusted p-values, and $\log_2$ fold changes.

Cell Type	Gene	$\log_2$ (Fold_Changes)	p_values	adj p_vals
Cancer Epithelial vs. other cell types	KRT8	4.78	< 0.001	< 0.001
Cancer Epithelial vs. other cell types	CLDN4	4.64	< 0.001	< 0.001
Cancer Epithelial vs. other cell types	FN1	4.99	< 0.001	< 0.001
Cancer Epithelial vs. other cell types	GNAS	5.58	< 0.001	< 0.001
Cancer Epithelial vs. other cell types	ZG16B	4.90	< 0.001	< 0.001
Cancer Epithelial vs. other cell types	AGR2	5.10	< 0.001	< 0.001
Cancer Epithelial vs. other cell types	FASN	4.01	< 0.001	< 0.001
Cancer Epithelial vs. other cell types	TFF3	5.02	< 0.001	< 0.001
Cancer Epithelial vs. other cell types	MUC1	5.159	< 0.001	< 0.001
Cancer Epithelial vs. other cell types	GPRC5A	4.05	< 0.001	< 0.001
Cancer Epithelial vs. other cell types	CALML5	5.50	< 0.001	< 0.001
Cancer Epithelial vs. other cell types	SCD	4.01	< 0.001	< 0.001
Cancer Epithelial vs. other cell types	REEP6	4.30	< 0.001	< 0.001
Cancer Epithelial vs. other cell types	MUCL1	4.14	< 0.001	< 0.001
Cancer Epithelial vs. other cell types	TNNT1	5.24	< 0.001	< 0.001
Cancer Epithelial vs. other cell types	PITX1	6.48	< 0.001	< 0.001
Cancer Epithelial vs. other cell types	CBLC	4.07	< 0.001	< 0.001
Cancer Epithelial vs. other cell types	PSCA	6.24	< 0.001	< 0.001
Cancer Epithelial vs. other cell types	TMEM184A	4.31	< 0.001	< 0.001
Cancer Epithelial vs. other cell types	PNMT	4.68	< 0.001	< 0.001

Supplementary Figures

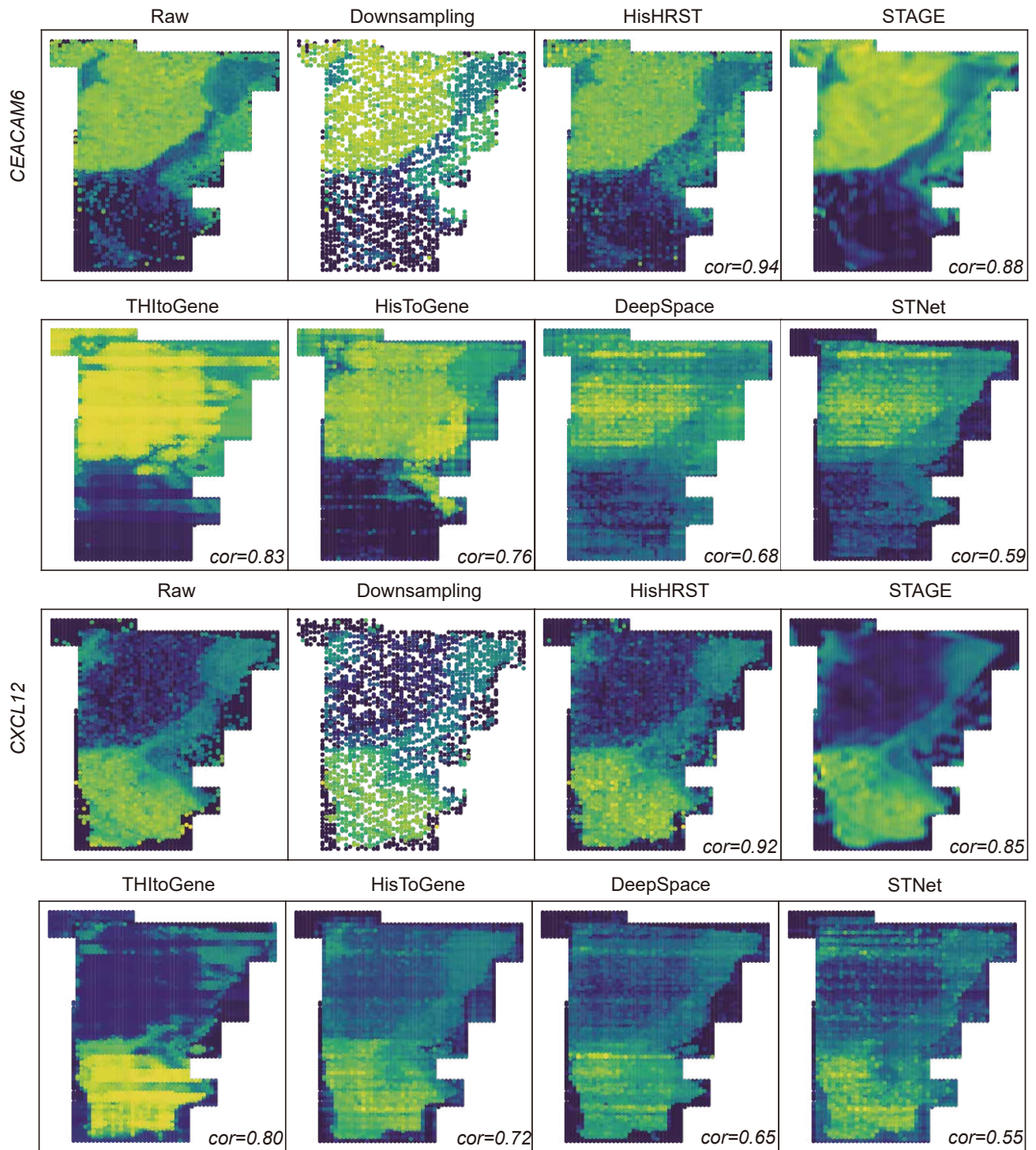


Figure S1. Visualizing the gene expression pattern reconstruction performance of HisHRST, STAGE, THltoGene, HisToGene, DeepSpaCE, and STNet on the marker genes *CEACAM6* and *CXCL12* in the TENX140 slice of the HEST-1k dataset.

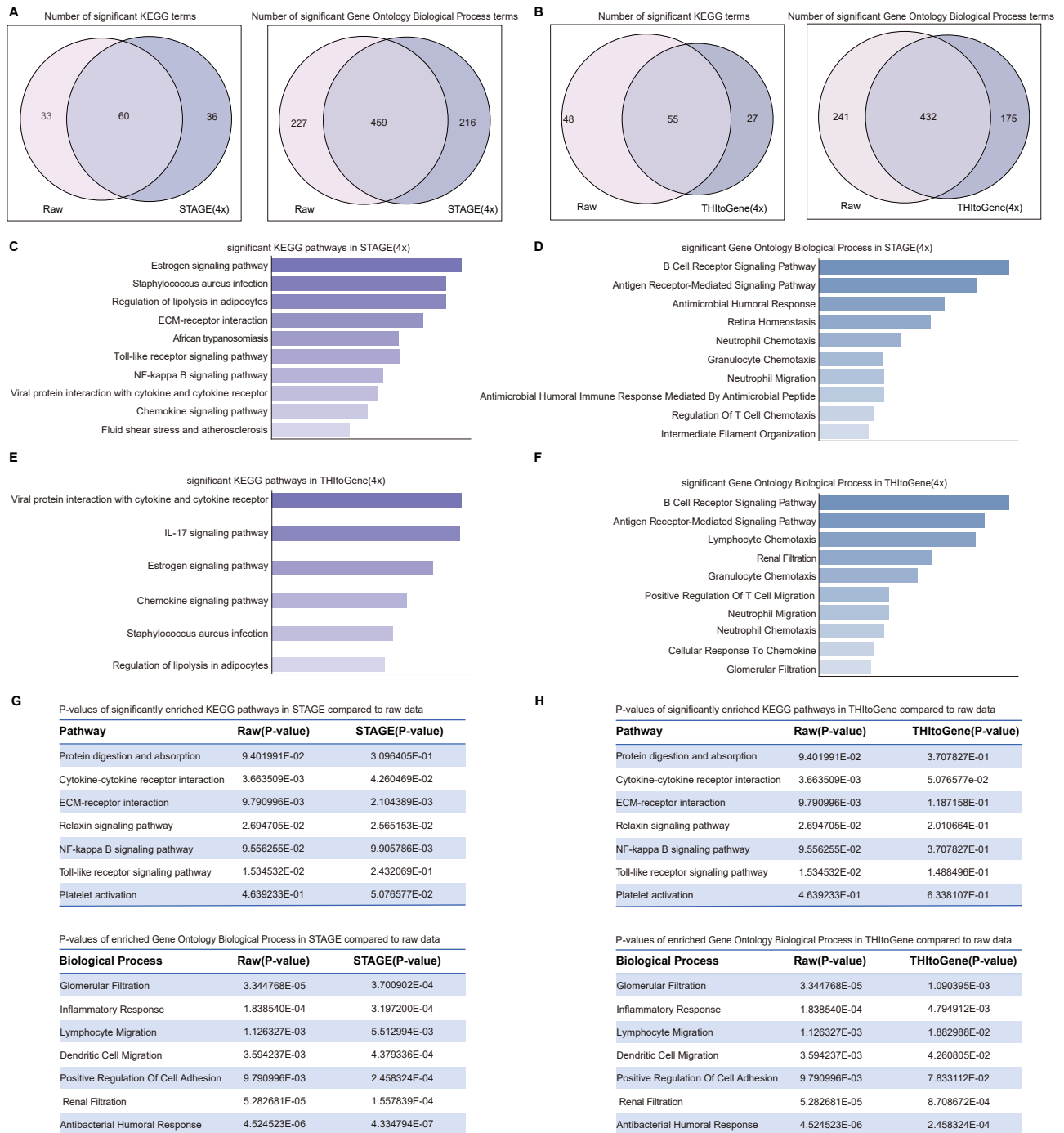


Figure S2. Biological functional analysis on the BC1 dataset. **(A)** The 4-fold data generated by STAGE enables the enrichment of a greater number of KEGG and Gene Ontology Biological Processes terms compared to the original data. **(B)** The 4-fold data generated by THIttoGene enables the enrichment of a greater number of KEGG and Gene Ontology Biological Processes terms compared to the original data. **(C and D)** Significantly enriched KEGG signaling pathways **(C)** and Gene Ontology Biological Processes **(D)** based on the 4-fold predicted data generated by STAGE. **(E and F)** Significantly enriched KEGG signaling pathways **(E)** and Gene Ontology Biological Processes **(F)** based on the 4-fold predicted data generated by THIttoGene. **(G and H)** Statistical significance of cancer-related signaling pathways and biological processes based on the 4-fold predicted data generated by STAGE and THIttoGene.

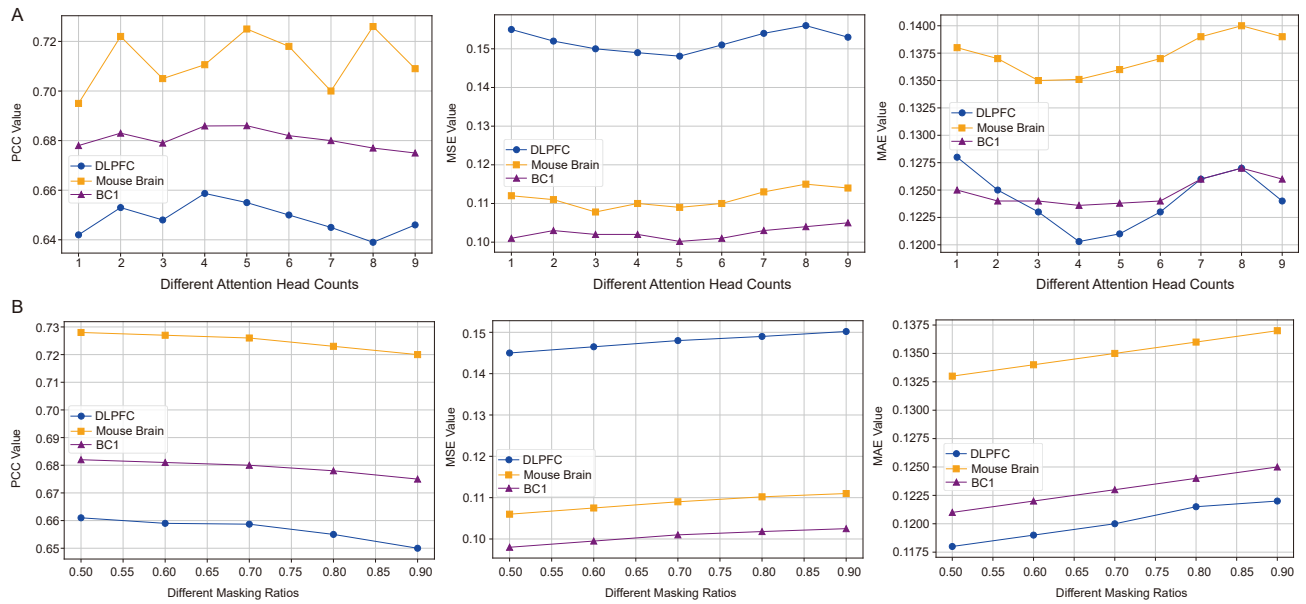


Figure S3. (A) Ablation study on the impact of different numbers of attention heads on performance across the DLPFC 151676 slice, MouseBrain dataset, and BC1 dataset. (B) Recovery performance under varying masking rates, simulating different levels of interpolation.

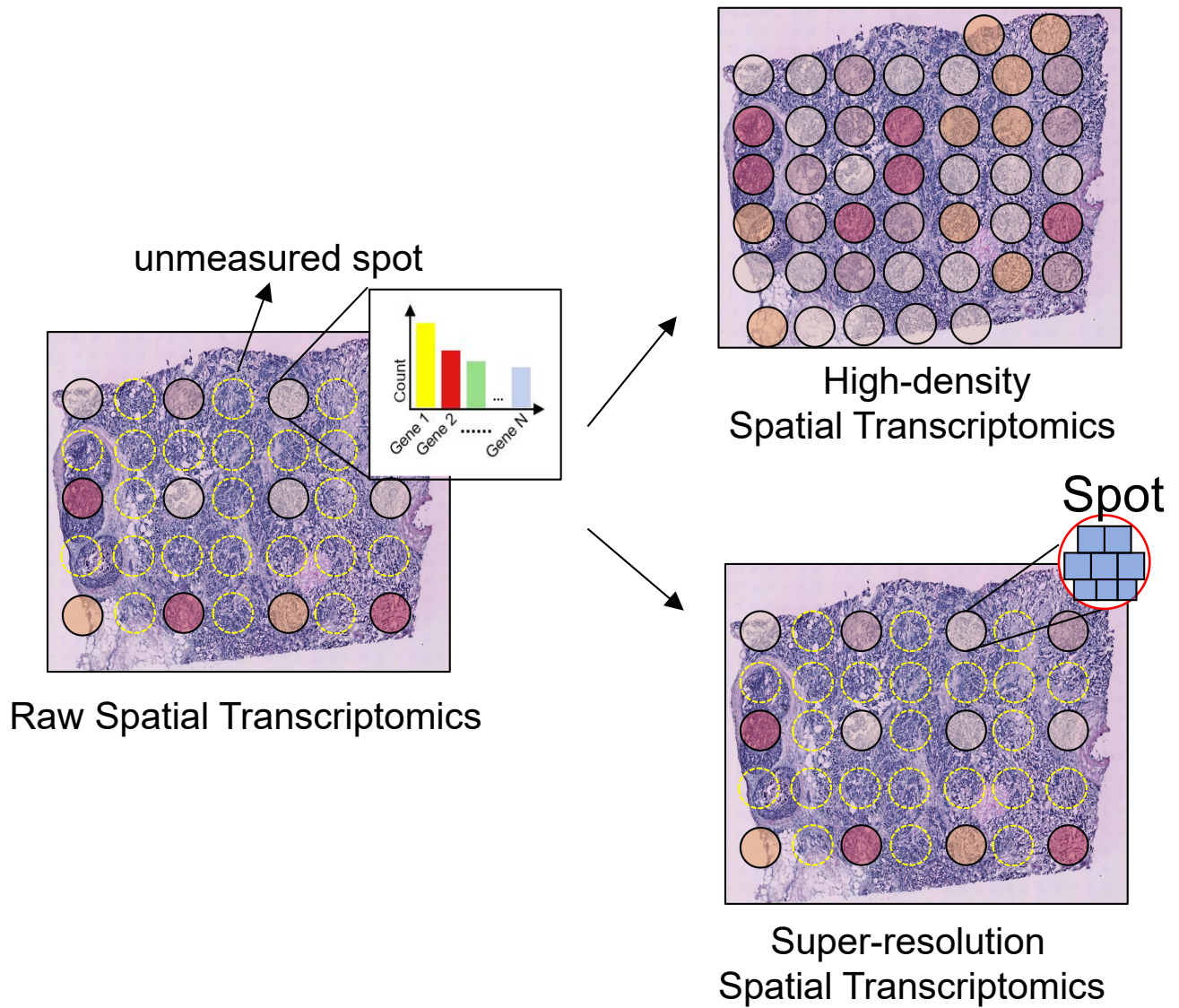


Figure S4. Comparison of generation paradigms between high-density spatial transcriptomics methods and super-resolution methods.

References

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4. Yuran Jia, Junliang Liu, Li Chen, Tianyi Zhao, and Yadong Wang. THItGene: a deep learning method for predicting spatial transcriptomics from histological images. *Briefings in Bioinformatics*, 25(1):464–474, 2024.
5. Shang Li, Kuo Gai, Kangning Dong, Yiyang Zhang, and Shihua Zhang. High-density generation of spatial transcriptomics with stage. *Nucleic Acids Research*, 52(9):4843–4856, 2024.